Cassava 7 — Known Issues and Gotchas

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The 90.32% headline is somewhat optimistic because the specialist was trained on different folds from the ones it was scored on. The notebooks also carry a few code bugs; check each against the version uploaded to Drive, which may already fix them.

Evaluation weak spots

Issue	Effect	What to do
Exp 11 specialist trained on different folds from the evaluation folds	Some evaluation images can have been in the specialist's training data, inflating 90.32%	Retrain the specialist on the same folds as the other ensemble members
The specialist loop reduces the learning rate on, and reports accuracy from, the same held-out fold	Mildly optimistic per-fold accuracy	Carve a validation slice from the training fold
Plain StratifiedKFold, not grouped	Safe for cassava (no duplicates found), but inconsistent with the cross-dataset protocol	Use StratifiedGroupKFold for new runs
Accuracy as the headline metric	Always guessing CMD scores 61.5%, so accuracy hides minority-class failure	Report balanced accuracy and macro-F1 alongside
One seed in the cross-dataset rerun (fold s.d. 0.56)	Uncertainty not fully measured	Add seeds 42 and 1337

Code bugs and weak spots

Where	Problem	Effect	Fix
Specialist training loop	Phase 2A unfreezes `model.layers[-30:]` — the last 30 layers of the whole model, not of the backbone	If the backbone is nested as one layer, it is fully unfrozen in 2A, not partly	Use a `get_backbone(model)` helper and unfreeze `backbone.layers[-30:]`; check how `build_specialist_model` nests the backbone
Data-pipeline cell	`mixup_batch` is defined twice with different signatures; the second replaces the first	`build_dataset_mixup` calls it with four arguments and will raise a TypeError	Rename one of them
Data-pipeline cell	`build_dataset(..., img_size='img_size')` defaults to the string `'img_size'`	`set_shape` fails when `augment=True` unless `img_size` is passed	Default to `CONFIG['img_size']`
Training loops	`if fold < N: continue` lines left from resuming interrupted runs	A fresh run silently skips folds	Reset before running
All pipelines	`preprocess_input` must run after augmentation	Before, colour augmentation acts on already-scaled values	Keep the current order

Environment errors already solved

Error	Cause	Fix
Saved weights would not reload (TF 2.19)	Old `.h5` save format	Save and load as `.keras`
Offline augmentation loop stopped early	The augmentation iterator ran out instead of repeating	Wrap it in `itertools.cycle`
Raspberry Pi: numpy ABI conflict	Camera stack and TensorFlow need different numpy builds	Run camera capture in a separate subprocess
Jetson: Pi Camera v2 fails on both CSI ports (I2C error -121)	Hardware or driver issue, unresolved	Time inference on pre-captured images
Kaggle session ends mid-run	Session time limit	Checkpoint per fold; save out-of-fold predictions after each fold